Exact Rank Obstruction and Dictionary-Relative Completion in a Finite Quadratic Response Model

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Abstract

We resolve an exact linear-algebra question in a fixed rational response model with 19 coordinates and 2,106 evaluation functionals. Nine native rows have rank 9; adjoining a distinguished target raises the rank to 10, whereas adjoining two formal-family rows gives rank 11. An explicit coordinate functional separates the target from the native coefficient span, while an exact identity locates it in the formal-family extension. A second, rank-13 assembly also misses the target. Five linear equalities and one nonvanishing condition classify every rational single-row addition that completes this assembly. The seven-element dictionary is a specialization: exactly r_5 and r_7 complete it, with both identities in closed form. A nineteen-texture minor has determinant $-2/9$, so evaluation separates the entire coefficient space. The classification is confined to this fixed response model and does not establish physical availability of its rows.

1 Introduction

Finite quadratic response models lead to a basic reachability problem: given a collection of response rows, does a distinguished target lie in their span, and if not, which additional generators remove the deficit? This note studies one such problem over $\mathbb{Q}$. The data consist of a 19-coordinate coefficient space, a pool of 2,106 scalar evaluations, nine native rows, a target, two formal-family rows, and a separate seven-element completion dictionary. The goal is not to infer the data from measurements, but to certify exactly what these fixed data can and cannot span.

The rows are the stipulated inputs of a finite generator-selection problem, not observations from an experiment. Their coordinates distinguish scalar, gradient, and quadratic response terms in the declared model. This makes the example a test of which extra response generators resolve a specified linear obstruction; it does not derive a constitutive law or assert that

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a particular apparatus realizes these rows. No physical interpretation is attached to these results.

Three features make the example useful. First, small rational minors certify a rank hierarchy $9 < 10 < 11$ without numerical conditioning questions. Second, an explicit coordinate functional identifies the direction missing from the native rows. Third, an explicit annihilator calculation classifies all rational single-row completions of a larger rank-13 assembly. The original seven-candidate calculation is a corollary; its successful formulas share a coefficient forced by the native obstruction. A full-rank nineteen-texture certificate makes coefficient-space and texture-response statements equivalent. The classification uses elementary quotient-space linear algebra; its content is the explicit criterion for these stipulated rows.

The measurement language is standard. Model-relative distinguishability and operator-system span are discussed by Heinosaari, Mazzarella, and Wolf, while Scott develops informationally complete operator frames and reconstruction formulas [7, 11]. Exact experimental design over a finite candidate set is likewise an established optimization setting [1]. Here these ideas are used only in their elementary finite-dimensional form: a nonzero $d \times d$ response minor proves that d selected scalar evaluations distinguish the specified d-dimensional row family.

All conclusions are relative to the definitions below. The continuum names attached to a few rows are mnemonic labels, not a microscopic derivation or a claim of physical co-realizability. Section 2 defines the model and the evaluation map; Sections 3–6 prove the rank, obstruction, and completion statements; the supplement supplies the exact minors, including a machine-readable full-model certificate and a bounded independent checker.

2 The fixed response model

Definition 2.1 (Coefficient space)**.** Let $V = \mathbb{Q}^{19}$ in the ordered basis

$$(a_0, a_1, a_2, b_1, b_2, b_3, g_1, g_2, r_1, r_2, r_3, r_4, r_5, r_6, r_7, m_0, m_1, m_2, \mu_0).$$

All omitted coordinates in the row definitions below are zero.

The nine native rows are

$$\begin{aligned}
e_0 &= a_0 + \tfrac{2}{3}b_2 + \tfrac{1}{3}g_2,\\
e_1 &= a_1 + b_1 - b_3 + \tfrac{1}{2}g_1 - \tfrac{1}{2}g_2,\\
e_2 &= a_2 + b_1 - \tfrac{2}{3}b_2 + b_3 + \tfrac{1}{2}g_1 + \tfrac{1}{6}g_2,\\
\beta_0 &= \tfrac{1}{3}r_1 - \tfrac{1}{3}r_3 - \tfrac{1}{3}r_4 - \tfrac{1}{3}r_5 + \tfrac{2}{3}r_6,\\
\beta_1 &= \tfrac{1}{2}r_2 + \tfrac{1}{2}r_5 - r_7,\\
\beta_2 &= \tfrac{2}{3}r_1 - \tfrac{1}{2}r_2 + \tfrac{1}{3}r_3 - \tfrac{2}{3}r_4 - \tfrac{1}{6}r_5 - \tfrac{2}{3}r_6 + r_7,\\
M_s &= 2m_0 + \tfrac{1}{2}m_1 + \tfrac{1}{2}m_2, \qquad M_t = \mu_0, \qquad \kappa = r_1 + r_2.
\end{aligned}$$

The two formal-family rows and target are

$$F_A = b_2 - r_7, \qquad F_B = -\tfrac{1}{4}r_2, \qquad t = -\tfrac{2}{3}b_2 - \tfrac{1}{9}r_2 + \tfrac{2}{3}r_7. \tag{1}$$

The texture pool is also fixed. Let $\omega \in \{0,1,2\}$, let k range over the nine integer triples listed in Supplement A.1, and let the twelve-component field vector have either one unit entry or two unit entries. The resulting $3 \cdot 9 \cdot (12 + \binom{12}{2}) = 2{,}106$ textures are ordered by ω, then by the displayed order of k, then by field: the twelve single-unit fields in coordinate order, followed by the 66 two-unit fields in lexicographic order of their occupied-position pairs. The exact contractions defining all nineteen response coordinates are collected in Supplement A.1; no floating-point arithmetic enters the proof.

Write $\mathcal{T}$ for this ordered texture pool. Each texture defines a linear functional on V, and together they define the evaluation map

$$\mathcal{E} : V \longrightarrow \mathbb{Q}^{\mathcal{T}}, \qquad v \longmapsto \big(v(\tau)\big)_{\tau \in \mathcal{T}}.$$

Thus a row has both a coefficient vector in V and a response vector $\mathcal{E}(v)$. Unless stated otherwise, ranks and residual counts below refer to response vectors on $\mathcal{T}$; identities such as Eq. (1) are identities in the coefficient space V. A nonzero response minor certifies coefficientwise independence as well, because any relation in V would remain a relation after applying $\mathcal{E}$.

Lemma 2.2 (Full-coefficient separation)**.** *The evaluation map $\mathcal{E}$ has rank 19 and is injective on V. Nineteen textures from the specified pool already separate every pair of coefficient rows. In particular, a linear identity holds on all textures if and only if it holds coefficientwise in V.*

Proof. In the ordered coordinate basis of V, the nineteen texture columns listed in Supplement A.2 form a 19×19 matrix with determinant $-2/9$. Its entries are given by the texture contractions and supplied as exact rational data. The minor gives rank at least 19; $\dim V = 19$ gives the upper bound and injectivity. □

3 Ranks 9, 10, and 11

Theorem 3.1 (Exact rank hierarchy)**.** *On the texture pool of Section 2,*

$$\begin{aligned} \operatorname{rank}\{e_0, e_1, e_2, \beta_0, \beta_1, \beta_2, M_s, M_t, \kappa\} &= 9, \\ \operatorname{rank}\{e_0, e_1, e_2, \beta_0, \beta_1, \beta_2, M_s, M_t, \kappa, t\} &= 10, \\ \operatorname{rank}\{e_0, e_1, e_2, \beta_0, \beta_1, \beta_2, M_s, M_t, \kappa, F_A, F_B\} &= 11. \end{aligned}$$

Proof. Supplement A.3 displays exact square response minors with determinants

$$\det R_9 = \frac{2}{9}, \qquad \det R_{10} = -\frac{5}{81}, \qquad \det R_{11} = -\frac{1}{36}.$$

They give the three lower bounds. The corresponding families contain exactly 9, 10, and 11 rows, which gives the matching upper bounds. □

Proposition 3.2 (Coordinate obstruction)**.** *The functional*

$$\mathcal{L} = -\tfrac{2}{3}a_0^* + \tfrac{2}{3}a_2^* + b_2^* \tag{2}$$

annihilates every native row and satisfies $\mathcal{L}(t) = -2/3$. Consequently t is not in the native coefficient span in V.

Proof. Substitution in the nine displayed native rows gives zero; substitution in Eq. (1) gives $-2/3$. □

Proposition 3.3 (Formal-family identity)**.** *Coefficientwise in V,*

$$t = -\frac{2}{3}F_A + \frac{4}{9}F_B. \tag{3}$$

Proof. The right side is $-(2/3)(b_2 - r_7) + (4/9)(-r_2/4)$, which is exactly Eq. (1). □

Remark 3.4 (Model-relative number of readings). Every texture supplies one scalar linear functional. No set of fewer than d scalar readings can be injective on a d-dimensional response space. The displayed nonzero minors show that selected sets of $d = 9, 10, 11$ readings attain the corresponding ranks for the row families above. This is a statement only about the fixed model and texture pool, not a lower bound for arbitrary experiments, encodings, models, or operator families.

4 A thirteen-row named assembly

Four coordinate rows are appended to the native family:

$$S_{r_4} = r_4, \qquad H_{b_1} = b_1, \qquad H_{b_3} = b_3, \qquad H_{\mathrm{dilat}} = b_2.$$

Let $\mathcal{A}$ denote the resulting thirteen-row assembly. It is a second span problem, not a revision of the native rank-9 family.

The labels recall familiar continuum ingredients: parity-even quadratic torsion models [6, 2], Oseen–Frank and saddle-splay terms [3, 10], Ericksen–Leslie couplings [8, 5], and generalized continua [9, 4]. In this note they serve only as mnemonic names for the four finite rows displayed above. No argument depends on a microscopic model, material interpretation, or physical co-realization.

We use one fixed residual convention in the completion calculation. With the textures in the specified order, exact left-to-right pivot reduction of a response vector against an ordered row family produces a remainder in $\mathbb{Q}^{\mathcal{T}}$; its *residual count* is the number of nonzero coordinates. In the table below, the generator residual reduces $\mathcal{E}(r_j)$ against $\mathcal{E}(\mathcal{A})$, while the target residual reduces $\mathcal{E}(t)$ against the assembly augmented by r_j.

Proposition 4.1 (Rank-13 assembly)**.** *The named assembly $\mathcal{A}$ has rank* 13*. The target is not in its span and leaves* 792 *nonzero texture residuals after exact pivot reduction.*

Proof. The exact 13×13 minor in Supplement A.4 has determinant $-1/9$. The family contains thirteen rows, giving the upper bound. Exact pivot back-substitution of the target followed by a scan of all 2,106 responses gives 792 nonzero residuals. □

5 Single-row completion in the fixed model

Write $A = \mathrm{span}(\mathcal{A})$ and $B = A + \mathrm{span}\{t\}$. For a coefficient row $x \in V$, define the five linear forms

$$\begin{aligned}
\phi_1(x) &= x_{g_1} - \tfrac{1}{2}(x_{a_1} + x_{a_2}),\\
\phi_2(x) &= x_{g_2} - \tfrac{1}{3}x_{a_0} + \tfrac{1}{2}x_{a_1} - \tfrac{1}{6}x_{a_2},\\
\phi_3(x) &= 2x_{r_3} + x_{r_6},\\
\phi_4(x) &= x_{m_1} - \tfrac{1}{4}x_{m_0},\\
\phi_5(x) &= x_{m_2} - \tfrac{1}{4}x_{m_0},
\end{aligned} \tag{4}$$

and the additional functional

$$s(x) = -2x_{r_3} + 2x_{r_5} + x_{r_7}. \tag{5}$$

Theorem 5.1 (All rational single-row completions)**.** *For any $x \in V$,*

$$t \in A + \mathrm{span}\{x\} \quad \Longleftrightarrow \quad \phi_1(x) = \cdots = \phi_5(x) = 0 \quad \textit{and} \quad s(x) \neq 0.$$

Equivalently, the completing rows are exactly $B \setminus A$. For every such row the coefficient of x is forced, and

$$t - \frac{2}{3s(x)}x \in A. \tag{6}$$

The same criterion holds for the corresponding texture-response vectors.

Proof. Substitution shows that all five ϕ_i vanish on the thirteen rows of $\mathcal{A}$ and on t. They are independent: their coefficients at g_1, g_2, r_6, m_1, m_2, respectively, form the 5×5 identity matrix. Their common kernel therefore has dimension 14. Substitution also gives $s|_A = 0$ and $s(t) = 2/3$. By Proposition 4.1, $\dim A = 13$, so $\dim B = 14$. Since $s(t) \neq 0$, the subspace $B \cap \ker s$ is a hyperplane of B containing A, hence equals A. Consequently,

$$B = \bigcap_{i=1}^{5} \ker \phi_i, \qquad A = B \cap \ker s.$$

If the stated conditions hold, $x \in B$ and the left side of Eq. (6) lies in $B \cap \ker s = A$. Conversely, if $t = a + cx$ with $a \in A$, then $c \neq 0$ because $t \notin A$. Thus $x \in B$ and applying s gives $2/3 = c\, s(x)$, proving both the conditions and the forced coefficient. Lemma 2.2 transfers the equivalence to response vectors. □

Both parts of the criterion are needed. The row r_1 satisfies the five equalities but has $s(r_1) = 0$; the row r_3 has $s(r_3) = -2 \neq 0$ but $\phi_3(r_3) = 2$. Neither completes the target. The theorem classifies algebraically available rows of V, not physically realizable generators.

6 Seven-candidate completion

The candidate dictionary is the seven-element set $\mathcal{D} = \{r_1, \ldots, r_7\}$. Each candidate is appended singly to $\mathcal{A}$. The following specialization of Theorem 5.1 retains the dictionary-specific identities and the complete residual enumeration.

Corollary 6.1 (Dictionary-relative completion)**.** *Exactly r_5 and r_7 complete the named assembly to the target. Their augmented ranks are* 14*, witnessed by determinants* $-2/9$ *and* $-1/9$*. The exact identities are*

$$t = -\tfrac{2}{9}\beta_0 - \tfrac{8}{9}\beta_1 - \tfrac{2}{9}\beta_2 + \tfrac{2}{9}\kappa - \tfrac{2}{9}S_{r_4} - \tfrac{2}{3}H_{\text{dilat}} + \tfrac{1}{3}r_5, \tag{7}$$

$$t = \tfrac{1}{9}\beta_0 + \tfrac{1}{9}\beta_1 + \tfrac{1}{9}\beta_2 - \tfrac{1}{9}\kappa + \tfrac{1}{9}S_{r_4} - \tfrac{2}{3}H_{\text{dilat}} + \tfrac{2}{3}r_7. \tag{8}$$

Proof. The rows r_1, r_2, r_4 fail the nonzero test in Theorem 5.1; r_3, r_6 fail its third equality. The rows r_5, r_7 satisfy all five equalities and have s-values $2, 1$, respectively, so they are exactly the completing dictionary members. For each r_j, exact pivot reduction determines whether the appended row raises rank and whether the target residual then vanishes on all 2,106 textures. The complete table is

candidate	augmented rank	generator residual	target residual	closes
r_1	13	0	792	no
r_2	13	0	792	no
r_3	14	234	756	no
r_4	13	0	792	no
r_5	14	792	0	yes
r_6	14	819	234	no
r_7	14	792	0	yes

Supplement A.4 displays both nonzero rank-14 minors. Back-substitution gives Eqs. (7) and (8); direct substitution kills all nineteen coefficient residuals and the full texture scan kills all 2,106 response residuals. □

Remark 6.2 (The obstructed direction). The functional $\mathcal{L}$ from Eq. (2) annihilates the nine native rows, S_{r_4}, H_{b_1}, H_{b_3}, and every candidate r_j, while $\mathcal{L}(H_{\text{dilat}}) = 1$. Hence any coefficient-space expansion of t using $\mathcal{A}$ and one candidate from $\mathcal{D}$ must give H_{dilat} the coefficient $\mathcal{L}(t) = -2/3$. The common coefficient in Eqs. (7) and (8) is therefore forced by the obstruction, rather than selected by the enumeration. Subtracting the two identities also shows that $r_5 - 2r_7 \in \text{span}(\mathcal{A})$: the successful candidates determine proportional nonzero directions in the quotient $V/\text{span}(\mathcal{A})$. This $-2/3$ coefficient law is dictionary-relative, not a statement for every $x \in V$. For example, the allowed algebraic choice $x = t$ completes with no assembly contribution. In a general expansion $t = a + cx$, the coefficient of H_{dilat} in a is $-2/3 - c\mathcal{L}(x)$; it equals $-2/3$ for the dictionary because $\mathcal{L}(r_j) = 0$.

7 Conclusion

The fixed response model has a simple exact structure. Its nine native rows have rank 9, the target adds one independent direction, and two formal rows contain that target through an explicit coefficient identity. A larger rank-13 assembly still misses the target. Five independent annihilator conditions and one nonzero test classify every rational single-row completion within V. Precisely two members of the seven-element dictionary pass. The native obstruction functional explains the coefficient shared by those two formulas, while

the general criterion identifies the required quotient direction. Nineteen specified textures separate the entire coefficient space, so this classification is equivalent in coefficient and response form.

All calculations use rational arithmetic. An independent implementation reconstructed the texture pool, response rows, ranks, obstruction values, residual counts, and completion identities directly from the definitions. The supplement records the rank certificates and provides the full-model minor as exact rational data. The accompanying bounded checker independently reconstructs that minor from the four-index projector formulas, verifies the annihilator criterion and dictionary identities, and tests negative controls. It does not replay the complete 2,106-texture residual counts.

The result is deliberately finite and model-relative. The classification covers all rational single-row additions in this 19-coordinate model, but does not classify generators in larger response spaces or arbitrary multi-row additions. It proves neither universal measurement minimality, stability under perturbation, nor a physical realization of the mnemonic continuum ingredients. Extension to larger response models requires their own row definitions and separation certificates.

A Exact response and certificate supplement

This supplement records the texture contractions and exact square minors used in the proofs, including the accompanying machine-readable full-model minor. Rows and columns occur in the orders stated in the text. All texture indices are zero-based.

A.1 Texture contractions

Order the field vector as

$$(w_0^0, w_1^0, w_2^0, w_{00}, w_{01}, w_{02}, w_{10}, w_{11}, w_{12}, w_{20}, w_{21}, w_{22}).$$

Let $\omega \in \{0, 1, 2\}$ and

$$k \in \{(0,0,0), (1,0,0), (0,1,0), (0,0,1), (1,1,0), (1,0,1), (0,1,1), (1,1,1), (1,2,3)\}.$$

The field vector has either one unit entry or two unit entries, yielding 78 fields and 2,106 textures. The order is ω first, then k in the displayed order, then the twelve single-unit fields in coordinate order followed by the 66 pairs in lexicographic order of their occupied positions. This is not lexicographic sorting of the resulting binary field vectors. Define

$$\tau = \sum_a w_{aa}, \quad |k|^2 = \sum_i k_i^2, \quad d_a = \sum_i k_i w_{ai}, \quad \ell_i = \sum_a k_a w_{ai}, \quad k \cdot w^0 = \sum_a k_a w_a^0.$$

With P_0, P_1, P_2 the trace, antisymmetric, and symmetric-traceless projectors,

$$\begin{aligned}
(P_0)_{ai,bj} &= \delta_{ai}\delta_{bj}/3, \\
(P_1)_{ai,bj} &= (\delta_{ab}\delta_{ij} - \delta_{aj}\delta_{ib})/2, \\
(P_2)_{ai,bj} &= (\delta_{ab}\delta_{ij} + \delta_{aj}\delta_{ib})/2 - \delta_{ai}\delta_{bj}/3.
\end{aligned}$$

The nineteen scalar responses are

$$
\begin{aligned}
a_s &= \omega^2 \langle w, P_s w \rangle && (s = 0, 1, 2), \\
b_1 &= \omega \sum_{ai} w_{ai} k_i w^0_a, && b_2 = \omega\tau (k \cdot w^0), \\
b_3 &= \omega \sum_{ai} w_{ai} k_a w^0_i, && g_1 = |k|^2 |w^0|^2, \\
g_2 &= (k \cdot w^0)^2, && r_1 = -|k|^2 \sum_{ai} w^2_{ai}, \\
r_2 &= -|k|^2 \tau^2, && r_3 = -|k|^2 \sum_{ai} w_{ai} w_{ia}, \\
r_4 &= -\sum_a d_a^2, && r_5 = -\sum_i \ell_i^2, \\
r_6 &= -\sum_a d_a \ell_a, && r_7 = -\left(\sum_{ai} k_a k_i w_{ai} \right) \tau, \\
m_s &= -\langle w, P_s w \rangle && (s = 0, 1, 2), \\
\mu_0 &= |w^0|^2.
\end{aligned}
$$

A.2 Full-coefficient separation certificate

Let $u_1, \ldots, u_{19}$ be the ordered coordinate basis of V from Section 2. Select the following texture indices, in this order:

$$
\begin{aligned}
J = (&0, 3, 4, 45, 78, 79, 81, 82, 84, 85, 86, 123, 149, \\
&705, 706, 747, 794, 798, 805).
\end{aligned}
$$

If j_m is the m-th entry of J, define

$$
M_{\ell m} = \mathcal{E}(u_\ell)_{j_m}, \qquad 1 \le \ell, m \le 19.
$$

Every entry is obtained from the contractions in Supplement A.1. Exact rational elimination gives

$$
\det M = -\frac{2}{9}.
$$

The file `data/full_model_separating_minor.csv`, relative to the package root, supplies all 361 entries. Its header records the ordered indices J; its first column names the coordinate rows. Integers and fractions are exact values, not rounded decimal measurements. The displayed formulas and these data specify every entry of the certificate.

The script `check_response.py` reconstructs all nineteen columns independently, using rational arithmetic and direct four-index projector contractions, compares every CSV entry, and checks rank 19 and determinant $-2/9$. It also verifies the five independent annihilators, $s|_A = 0$, $s(t) = 2/3$, and the dictionary specialization. Its negative controls include a row with $s \neq 0$ that violates an annihilator, a row in the assembly span with $s = 0$, a deficient texture selection, and corrupted certificate data. The checker is bounded: it does not replay the full residual enumeration or all six smaller displayed minors. It uses only the Python standard library; the README gives the command and the output boundary.

A.3 Exact rank 9, 10, and 11 minors

Column integer j denotes texture index j in the specified zero-based ordered construction. All entries below are rational numbers.

Rank 9. The rows are

$$e_0, e_1, e_2, \beta_0, \beta_1, \beta_2, \qquad M_s, M_t, \kappa.$$

The columns are $0, 3, 78, 79, 81, 82, 86, 149, 747$.

$$\begin{pmatrix} 0 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 4/3 \\ 0 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2/3 & 1/2 & 0 & 0 & 0 & 0 & 2/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1/3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1/2 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1/2 & -2/3 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & -1/2 & -1/2 & -3 & -3 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & -1 & -1 & -6 & 0 \end{pmatrix}$$

Its determinant is $2/9$.

Rank 10. Append the target and use columns $0, 3, 78, 79, 81, 82, 84, 86, 149, 747$.

$$\begin{pmatrix} 0 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 & 4/3 \\ 0 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2/3 & 1/2 & 0 & 0 & 0 & 0 & 0 & 2/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1/3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1/2 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1/2 & 0 & -2/3 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 & -1/2 & -1/2 & -1/2 & -3 & -3 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & -1 & -1 & -1 & -6 & 0 \\ 0 & 0 & 0 & 0 & -5/9 & 0 & 0 & 0 & 4/9 & 0 \end{pmatrix}$$

Its determinant is $-5/81$.

Rank 11. Append F_A, F_B and use columns $0, 3, 78, 79, 81, 82, 84, 85, 86, 149, 747$.

$$\begin{pmatrix}
0 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4/3 \\
0 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 2/3 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 2/3 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1/3 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1/2 & 0 & -1/2 & 0 & -2 & 0 \\
0 & 0 & 0 & 0 & 0 & -1/2 & 0 & -1/2 & -2/3 & 0 & 0 \\
0 & -1 & 0 & 0 & -1 & -1/2 & -1/2 & -1 & -1/2 & -3 & -3 \\
1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -2 & -1 & -1 & -2 & -1 & -6 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1/4 & 0 & 0 & 1/4 & 0 & 1 & 0
\end{pmatrix}$$

Its determinant is $-1/36$.

A.4 Rank 13 and completion minors

The rank-13 rows are $e_0, e_1, e_2, \beta_0, \beta_1, \beta_2, M_s, M_t, \kappa, S_{r_4}, H_{b_1}, H_{b_3}, H_{\mathrm{dilat}}$ and the columns are $0, 3, 78, 79, 81, 82, 84, 86, 149, 747, 794, 798, 805$.

$$\begin{pmatrix}
0 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 & 4/3 & 4/3 & 4/3 & 0 \\
0 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 2/3 & 1/2 & 0 & 0 & 0 & 0 & 0 & 2/3 & 8/3 & 2/3 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1/3 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1/2 & 0 & 0 & -2 & 0 & 0 & -1/2 & -1/2 \\
0 & 0 & 0 & 0 & 0 & -1/2 & 0 & -2/3 & 0 & 0 & 0 & -1/2 & -1/2 \\
0 & -1 & 0 & 0 & -1 & -1/2 & -1/2 & -1/2 & -3 & -3 & -1 & -1 & -1/2 \\
1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & -2 & -1 & -1 & -1 & -6 & 0 & -2 & -2 & -1 \\
0 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0
\end{pmatrix}$$

Its determinant is $-1/9$.

For r_5, append its response row and use the columns

$$0, 3, 78, 79, 81, 82, 84, 85, 86, 149, 747, 794, 798, 805.$$

$$\begin{pmatrix}
0 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4/3 & 4/3 & 4/3 & 0 \\
0 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 2/3 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 2/3 & 8/3 & 2/3 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1/3 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1/2 & 0 & -1/2 & 0 & -2 & 0 & 0 & -1/2 & -1/2 \\
0 & 0 & 0 & 0 & 0 & -1/2 & 0 & -1/2 & -2/3 & 0 & 0 & 0 & -1/2 & -1/2 \\
0 & -1 & 0 & 0 & -1 & -1/2 & -1/2 & -1 & -1/2 & -3 & -3 & -1 & -1 & -1/2 \\
1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & -2 & -1 & -1 & -2 & -1 & -6 & 0 & -2 & -2 & -1 \\
0 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1
\end{pmatrix}$$

Its determinant is $-2/9$.

For r_7, use the same columns and append the r_7 response row.

$$\begin{pmatrix}
0 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4/3 & 4/3 & 4/3 & 0 \\
0 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 2/3 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 2/3 & 8/3 & 2/3 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1/3 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1/2 & 0 & -1/2 & 0 & -2 & 0 & 0 & -1/2 & -1/2 \\
0 & 0 & 0 & 0 & 0 & -1/2 & 0 & -1/2 & -2/3 & 0 & 0 & 0 & -1/2 & -1/2 \\
0 & -1 & 0 & 0 & -1 & -1/2 & -1/2 & -1 & -1/2 & -3 & -3 & -1 & -1 & -1/2 \\
1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & -2 & -1 & -1 & -2 & -1 & -6 & 0 & -2 & -2 & -1 \\
0 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0
\end{pmatrix}$$

Its determinant is $-1/9$.

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